

Geospatial Technology

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Challenge Interpretation

Tectonic plates are large, solid slabs of rock that fit together like a puzzle to make up the Earth's crust. The Pacific Plate, the largest and fastest-moving tectonic plate, is home to the Ring of Fire, a 40,000 km-long belt characterized by its active volcanoes and frequent earthquakes. This horseshoe-shaped area surrounding the Pacific Ocean is responsible for most of the earthquakes that happen on Earth.

Our community, King County, is one of many areas situated in the Ring of Fire. The county is located within the Puget Sound region, which lies either near or on top of several faults, such as the Seattle Fault, Tacoma Fault, and the Cascadia Subduction Zone. It is also flanked by active volcanoes like Mount Rainier and Mount Baker. King County's history has been dotted with many significant earthquakes, including the magnitude 6.8 Nisqually earthquake that caused widespread damage in 2001. Because of its fragile location with many faults, people need to know the importance of tracking these earthquakes and their dangerous aftermaths.

As residents of the most populous city in Washington, the people of Seattle specifically face a greater risk of injury or death in the event of a major earthquake. Because of this, our team decided to focus our research and analyze the causes and impacts of seismic activity in Seattle. We have used GIS maps to overlap fault lines, population density, and critical infrastructure such as schools, hospitals, and major roads to identify high-risk areas. Precise data like studies of earthquake magnitude records, infrastructure sturdiness, and floodplain regions have been presented to show how different parts of the city may be affected.

Using data from the Washington Geographic Information Portal, we will then analyze a Seattle seismic scenario to predict where the most building and road damage will occur, how emergency services will be affected, and what the people of Seattle can do to stay safe. Our plans aim to find solutions that optimize evacuation strategies, reduce delays, and ensure the safety of Seattle residents during and after future seismic events.

Challenge Explanation

The 2024-2025 theme for Geospatial Technology is "Identify a disaster threat, natural or otherwise, that may impact your community."

According to the United Nations Office for Disaster Risk Reduction, a disaster risk is defined as "the potential loss of life, injury, or damaged assets which could occur to a system, society, or a community in a specific period of time..." (UNDRR). While disasters can be both man-made and natural, our team decided to focus on the latter. This is because the effects of natural disasters can be clearly seen through geospatial data.

Natural disasters in the U.S. caused over \$165 billion in damages in 2022 alone, and that number is only expected to rise. In the past five years, the country has faced countless catastrophic events like the devastating LA wildfires, the 2020 earthquake in Utah, Hurricane Ian tearing through Florida, the deadly COVID-19 pandemic, and record-breaking floods in Kentucky. These current disasters and their devastating impacts show how vulnerable our communities are, which is why a proper analysis needs to happen of their impacts to help with the readiness of our communities in the future.

This event requires our team to map danger zones, predict future risks, and develop potential solutions like evacuation routes and resource distribution centers. Geospatial data is essential for a proper analysis like this because it provides a spatial context, making it easier for users to find patterns, trends, and relationships between specific locations and data points on Earth.

TSA's Geospatial Technology also requires us to create an infographic that will clearly communicate all our team's research and data. We will use the power of visualizations to have people understand and be prepared for an emergency in case our researched disaster threat occurs.

Geospatial Technology helps teams practice leadership and several 21st-century skills like Communication, Initiative, Problem Solving, Critical Thinking, Perseverance/Grit, Creativity, and Flexibility. This event is also linked to future careers like being a GIS Analyst, Geospatial Programmer, Cartographer, Land Surveyor, Mapping Technician, and more.

Collected Geospatial Data

Historical Data + Causes of Earthquakes

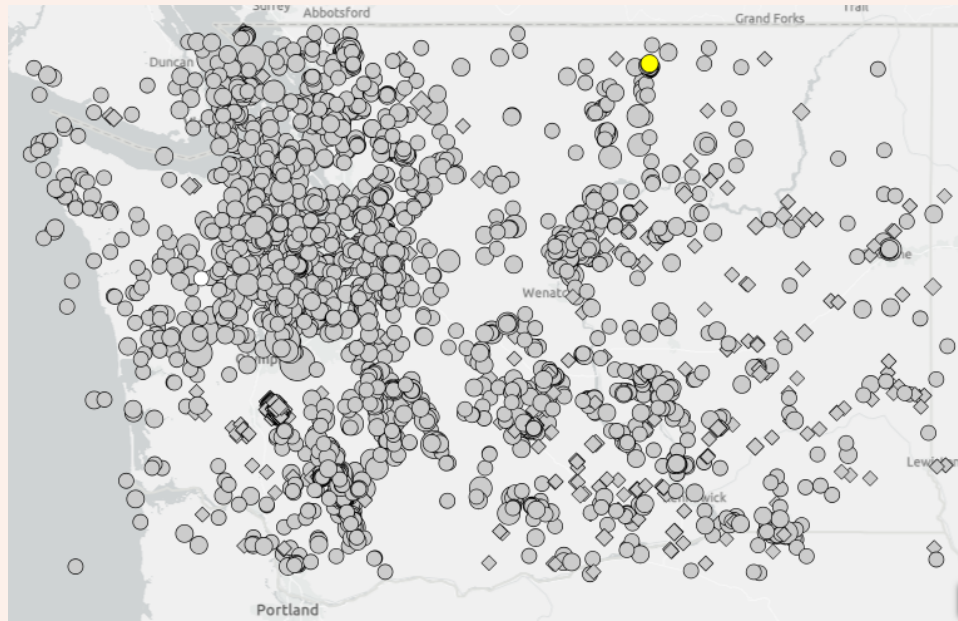


Figure 1: 6,534 WA Historical Earthquakes with Magnitudes > 2.5 since 1900 | USGS Earthquake Catalog

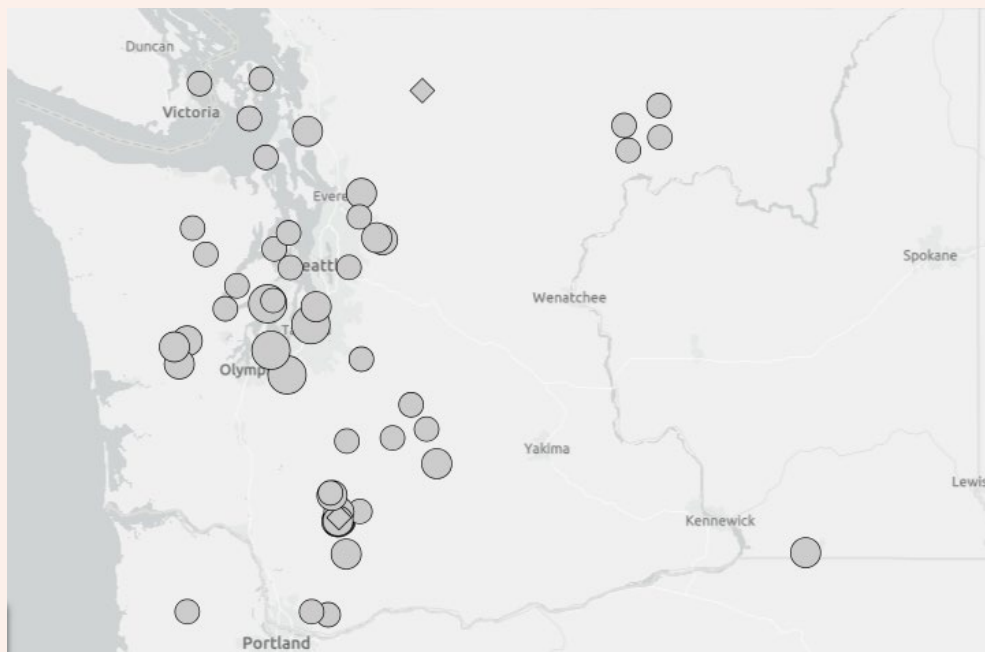


Figure 2: 137 WA Historical Earthquakes with Magnitudes > 4.5 since 1900 | USGS Earthquake Catalog

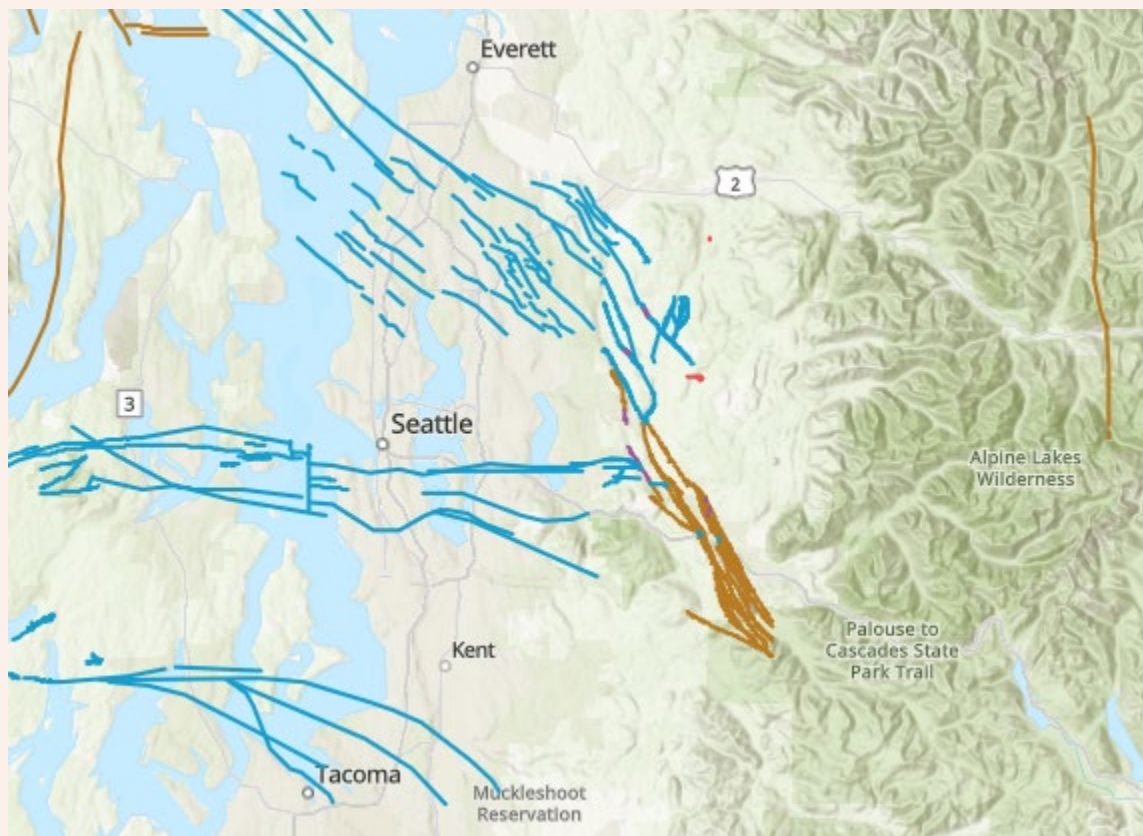


Figure 3: Fault Lines Through and Around Seattle | Esri U.S. Federal Datasets

Risks After Earthquakes - Flood Prone Areas

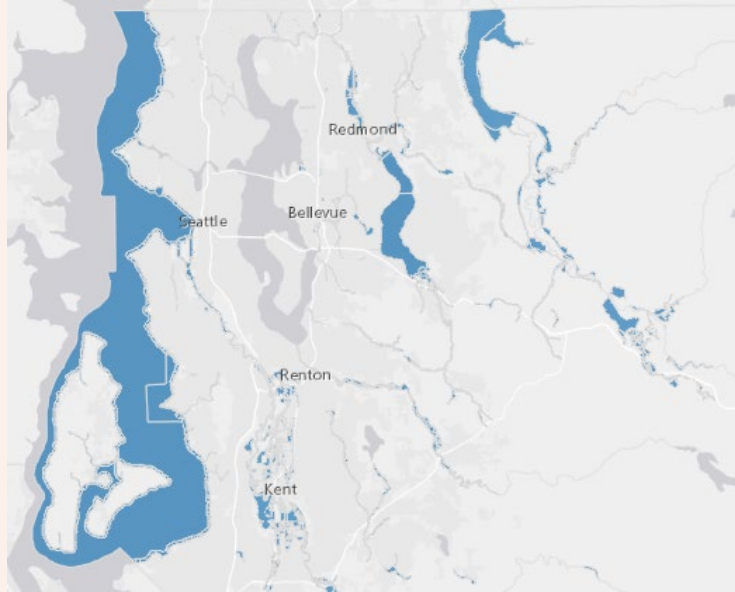


Figure 4: FEMA High Risk Flood Area (100-year Floodplain) | King County GIS Open Data

Risks After Earthquakes - Liquefaction Prone Areas

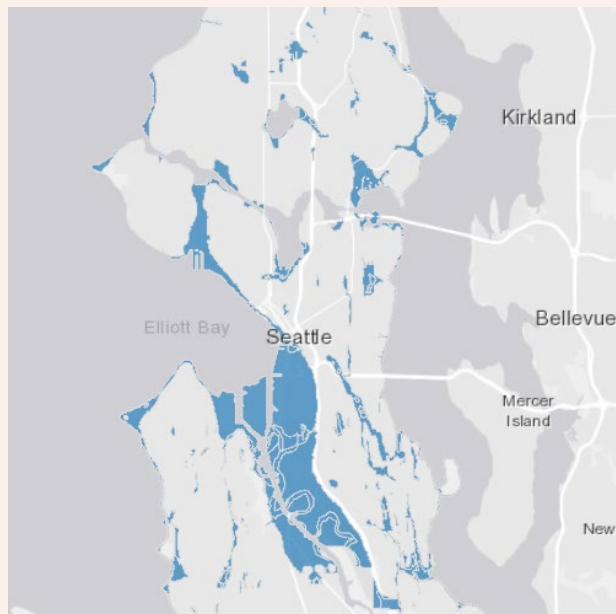


Figure 5: ECA Liquefaction Prone Areas | Seattle GeoData

Area	Acres	% of Seattle	% of Zone
Liquefaction Prone Areas	8029.46	15%	100%
Property in Areas	6172.02	12%	77%
Commercial/Mixed-Use -	718.78	1%	9%
Easement	2.07	0%	0%
Industrial	1510.42	3%	19%
Major Institution and Public Facilities/Utilities	2024.07	4%	25%
Multi-Family	217.96	0%	3%
Parks/Open Space/Cemeteries	463.74	1%	6%
Reservoirs/Water Bodies	2.92	0%	0%
Single Family	490.97	1%	6%
Unknown	17.31	0%	0%
Vacant	723.76	1%	9%
Right of Way in Areas	1857.44	3%	23%

Figure 6: Distribution of Land Use and Property Types in Liquefaction-Prone Areas of Seattle | Seattle Geodata

Risks After Earthquakes - Potential Structural Damage

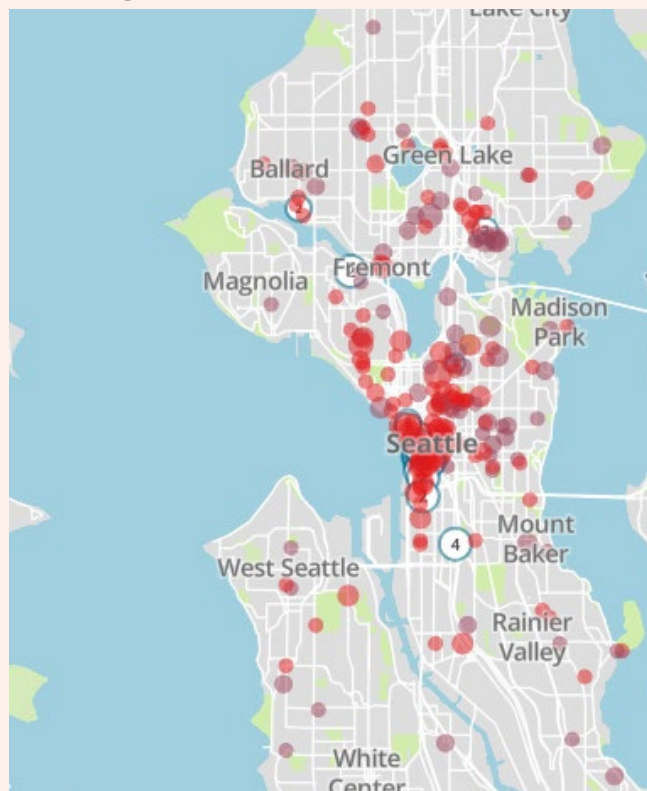


Figure 7: Seattle's 1160 Unreinforced Masonry Buildings | Seattle GIS Open Data

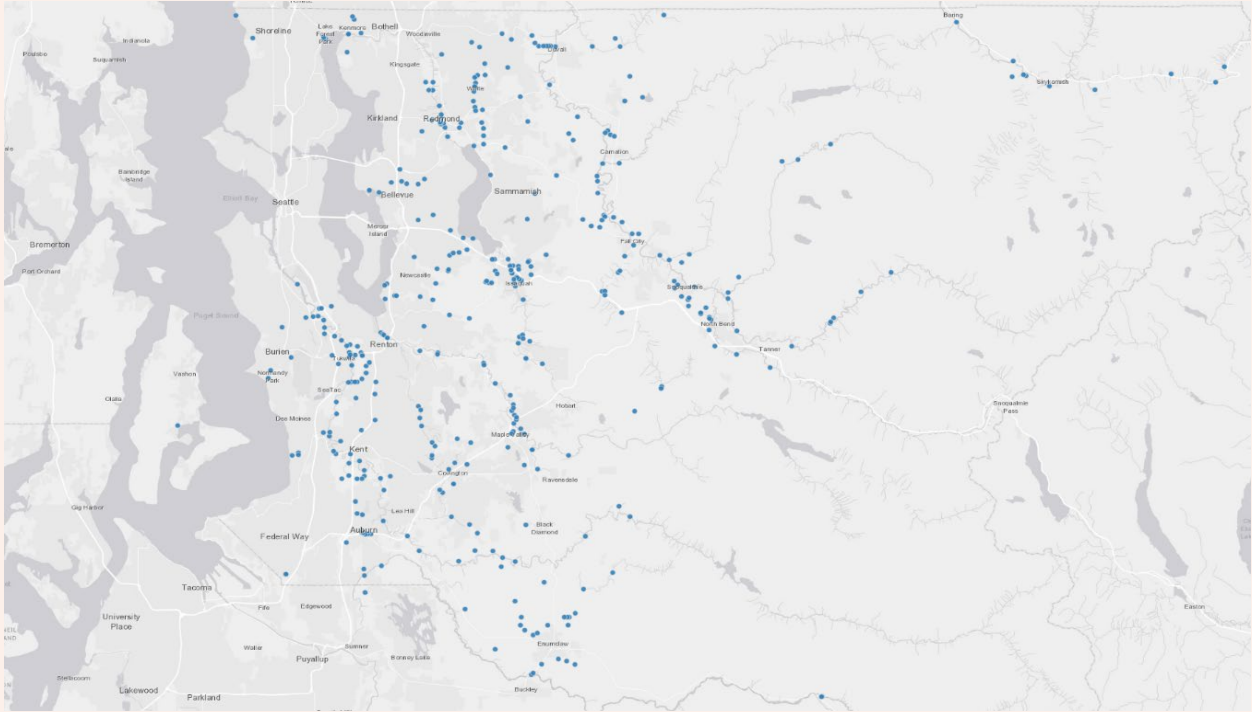


Figure 8: Bridges Maintained by King County | King County GIS Open Data

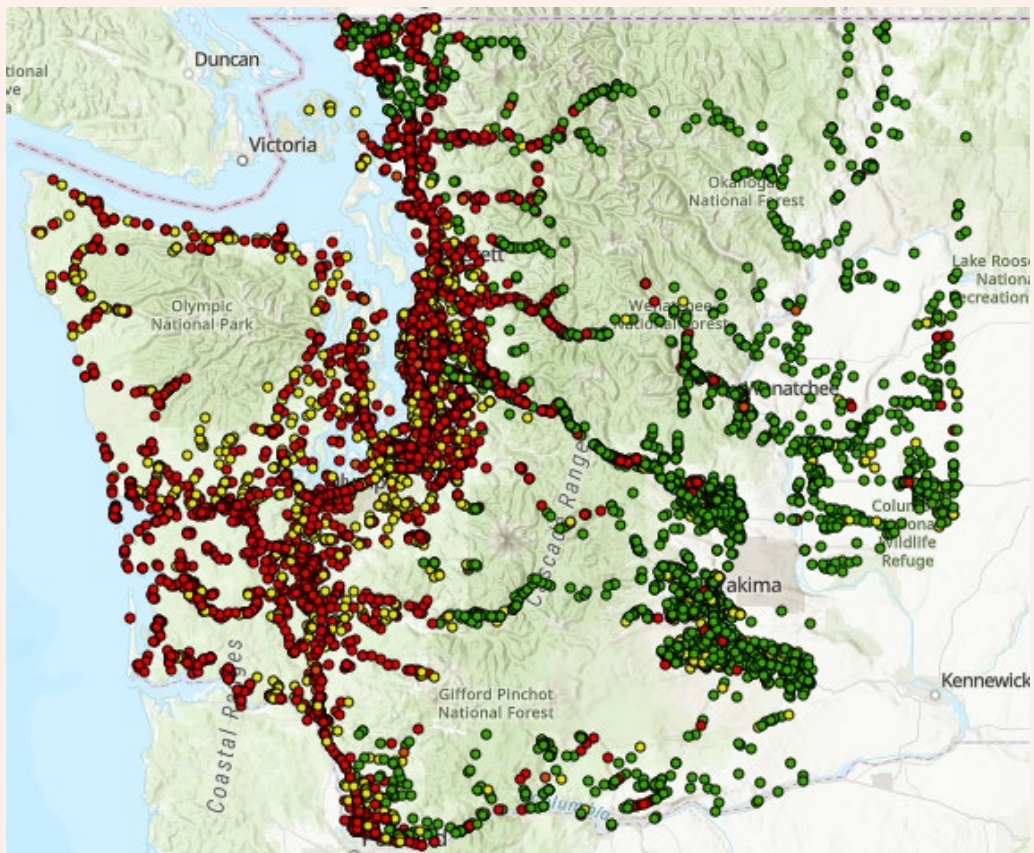


Figure 9: Seismic Bridge Resiliency Assessment | Timmons User

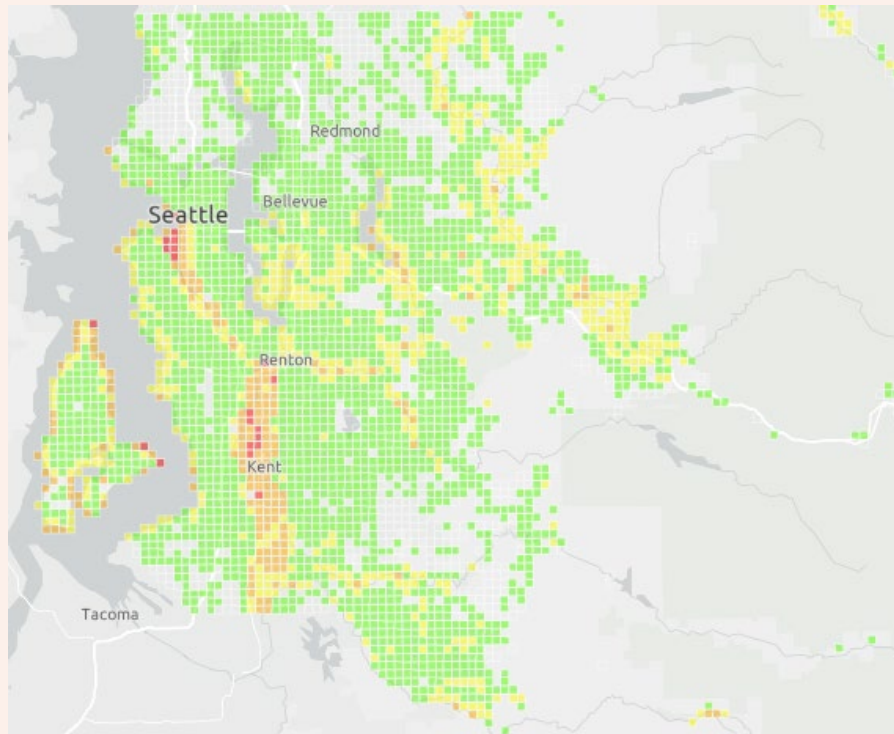


Figure 10: Average Number of Hazard Risks per Structure | FEME, King County

Others – Miscellaneous

Used Census 2020 blocks in King County with selected P.L. 94-171 redistricting data. for Population Density Map, no image shown because there were too many records (around 27,000) to show without zooming in.

<https://www.arcgis.com/home/item.html?id=033d9a0ec26f49339540c0e41481735f>

<https://www.seattle.gov/emergency-management/prepare/prepare-your-neighborhood/community-emergency-hubs> - Locations of all of Seattle's Emergency Community Hubs

Type	Size	Estimated Frequency
Deep Earthquakes like Nisqually	Over 6.0	Every 30 to 50 years
Megathrust Earthquakes	8.0 to 9.0+	Every 500 years on average
Seattle Fault	Over 5.5	Every 200 – 12,000 years
Seattle Fault	Over 7.2	Every 5000 – 15,000 years

Figure 11: Estimated Frequency of Earthquakes at Different Magnitudes

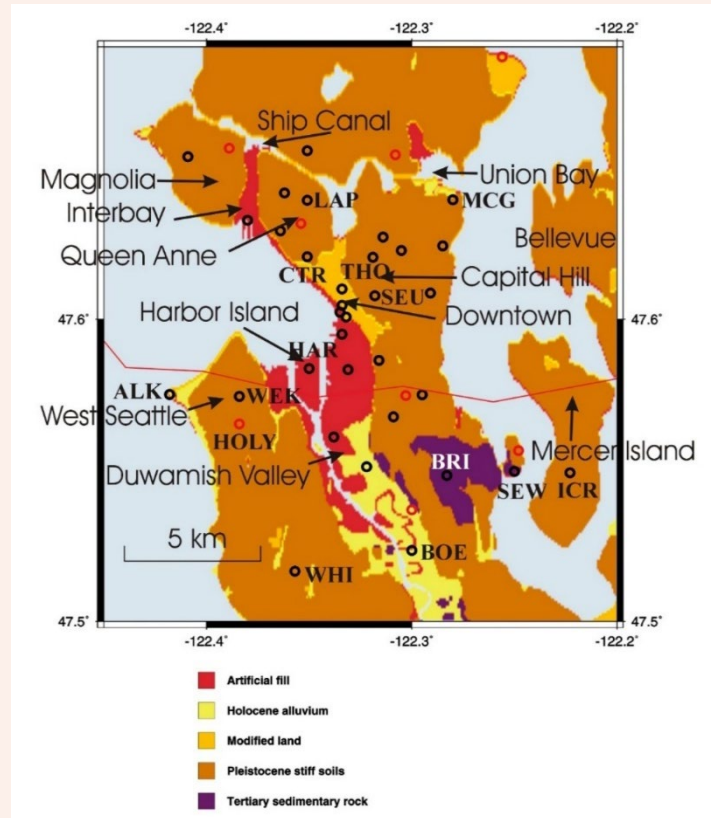


Figure 12: Estimated Frequency of Earthquakes at Different Magnitudes | Frankel, A - ResearchGate

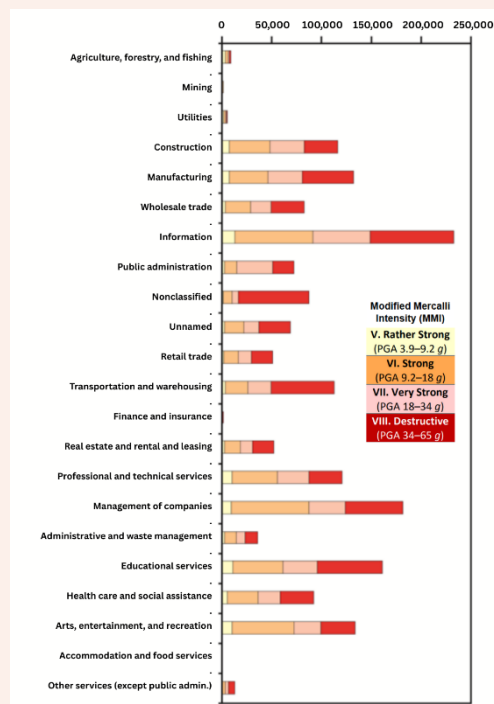


Figure 13: Projected Employee Impact Number by Sector in a Seattle M7.2 Earthquake: Distribution of Workforce Exposure to Different Intensity Levels | Seattle GeoData



Figure 14: Polygon File of City of Seattle Boundary | Seattle GIS Open Data

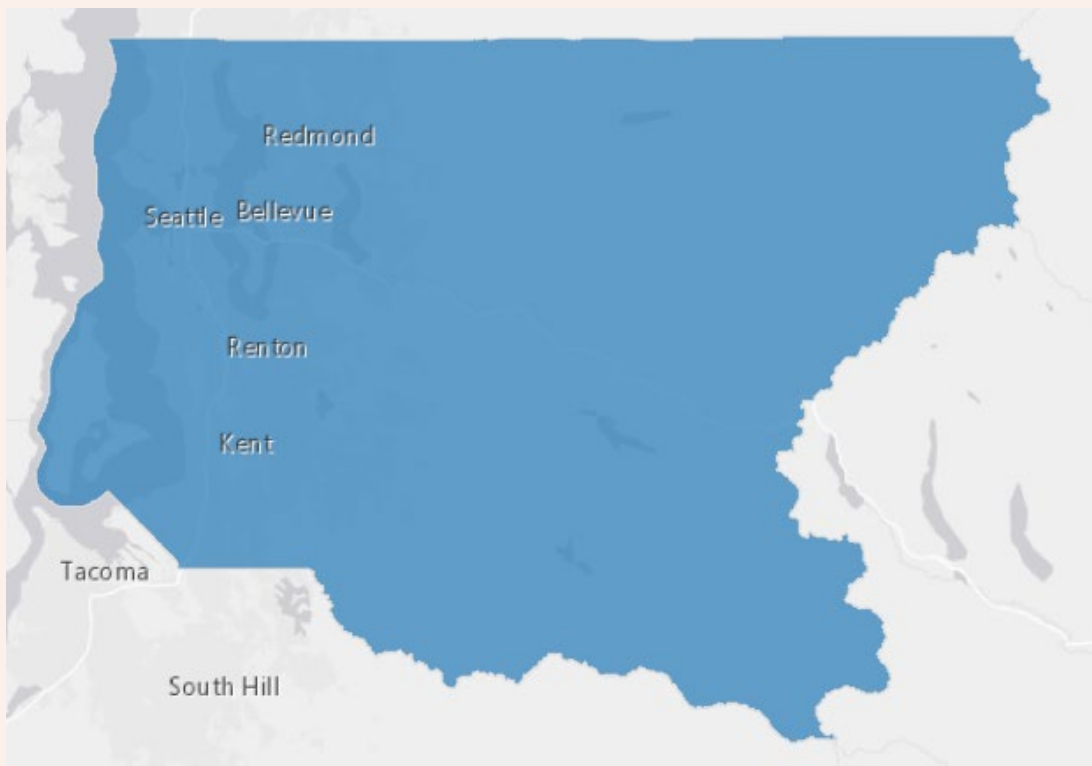


Figure 15: King County Political Boundary (no waterbodies) / kingco area | King County GIS Open Data

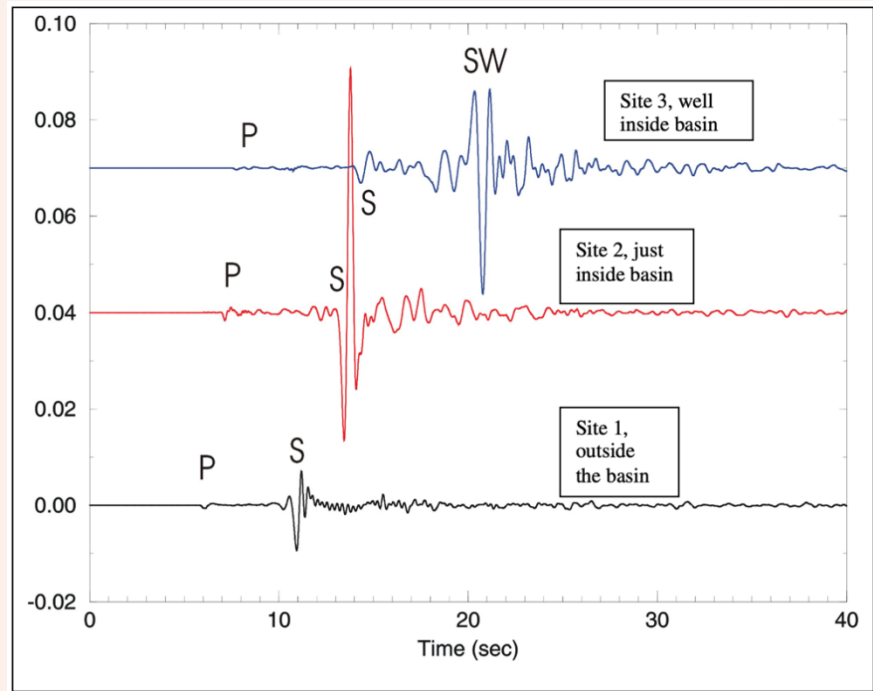


Figure 16: 3D Simulation East-West Velocity Seismograms | USGS

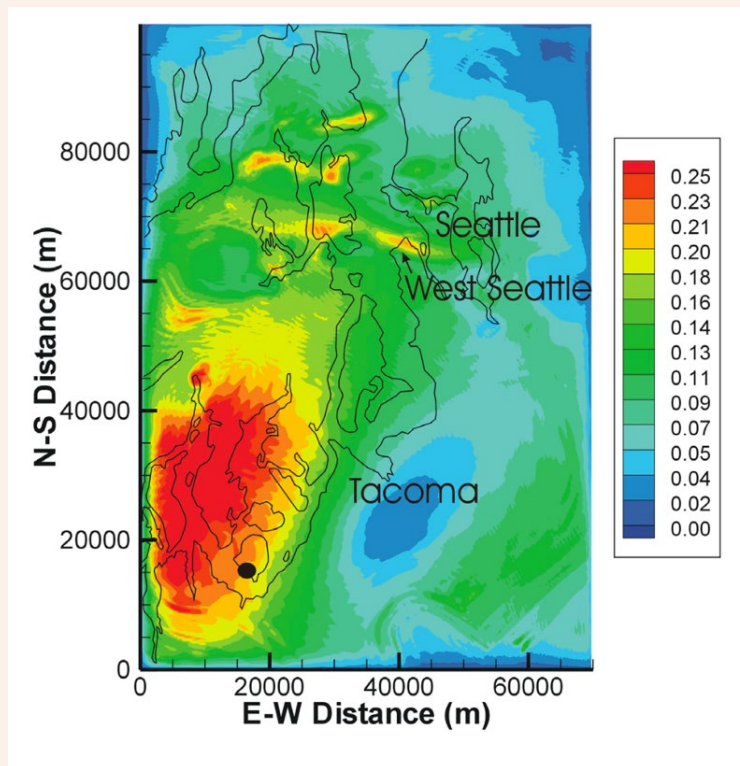


Figure 17: Peak Values of Root Mean Square Ground Velocity of Two Horizontal Components from 3D Simulation of 2001 Nisqually Earthquake | USGS

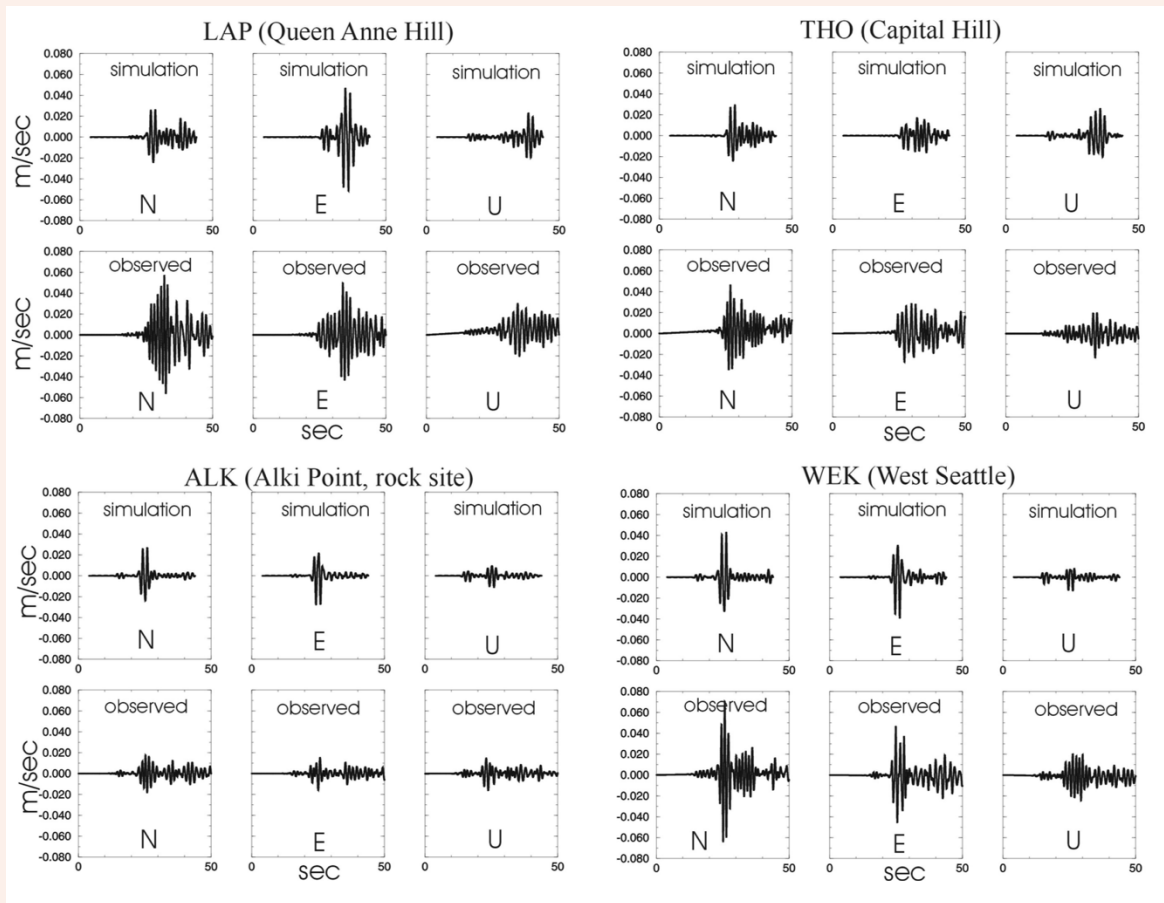


Figure 18: Synthetic and Observed Velocity Waveforms for 2001 M6.8 Nisqually Earthquake | USGS

<https://data-seattlecitygis.opendata.arcgis.com/datasets/hospitals-2/about> -

Provided Data for all the Hospitals in the Seattle Area

<https://gis-kingcounty.opendata.arcgis.com/datasets/kingcounty::police-station-locations-in-king-county-kcp-loc-point/about> - Police Station Locations in King

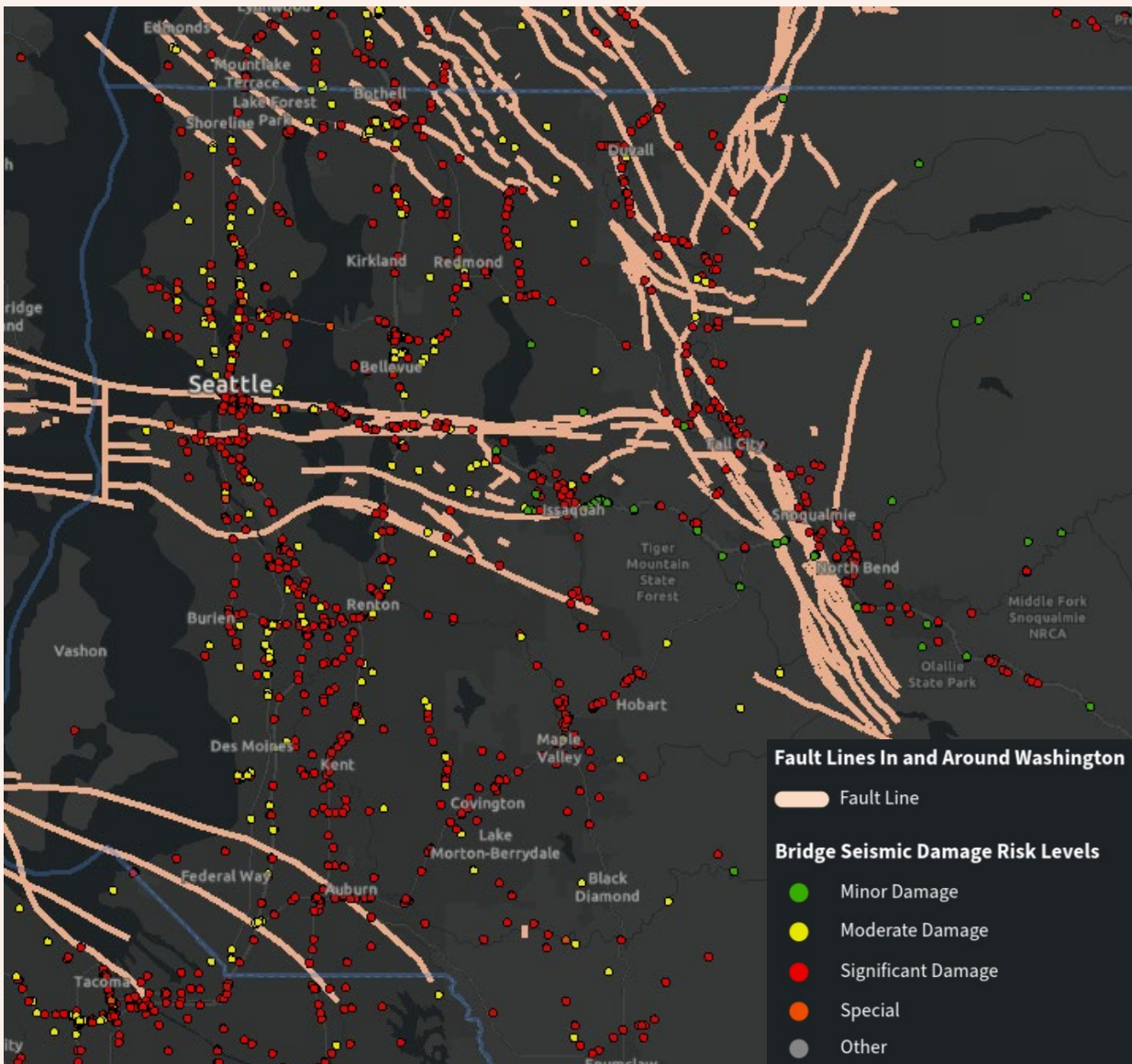
County

<https://gis-kingcounty.opendata.arcgis.com/datasets/kingcounty::fire-station-locations-in-king-county-firestn-point/about> - Fire Station Locations in King County

Analysis Maps

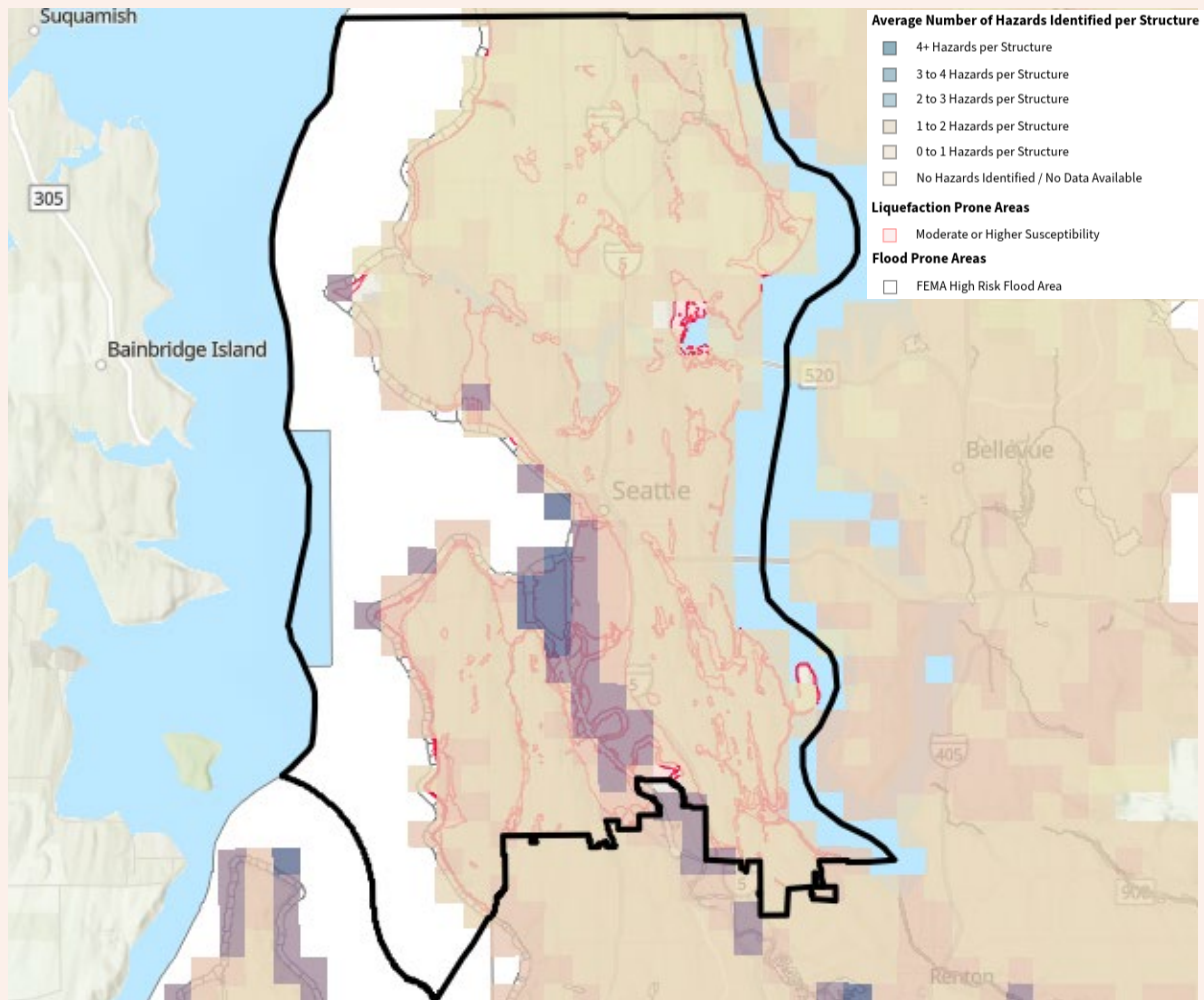
Analysis Map #1:

Overlay of King County Seismic Bridge Risk Locations and Fault Lines in and Around Washington



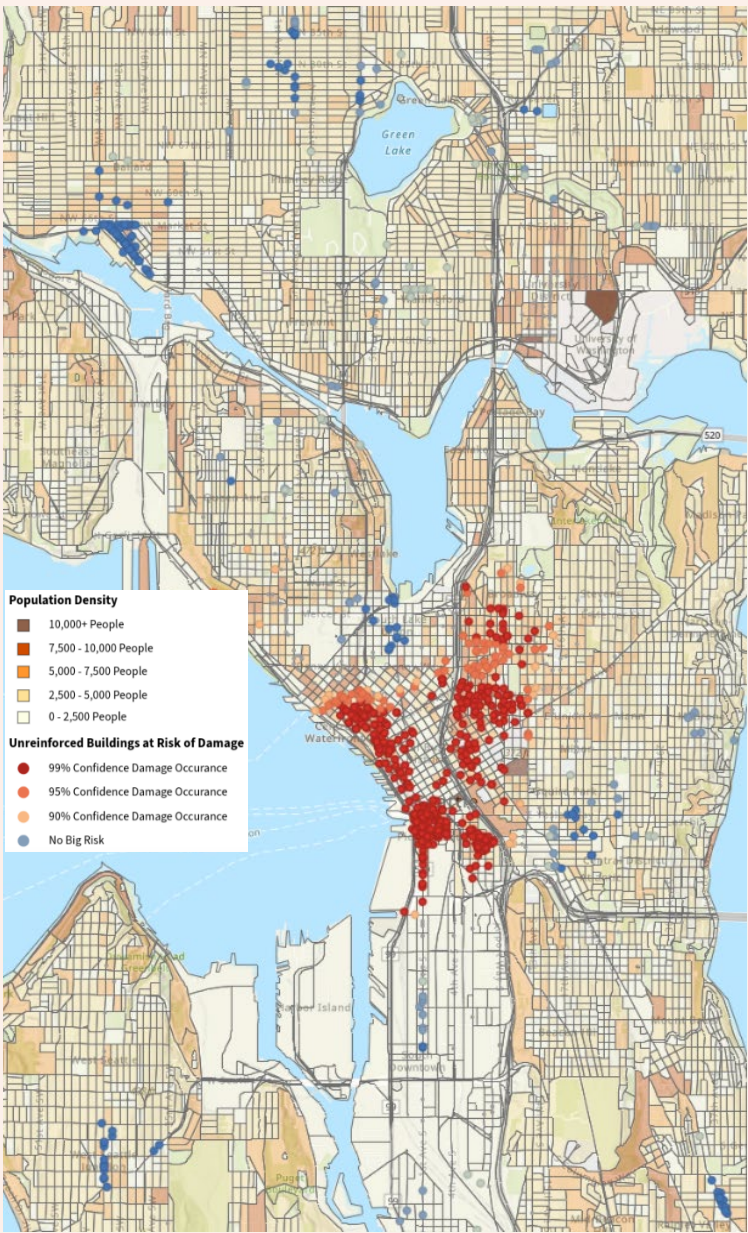
Analysis Map #2:

Liquefaction & Flood Prone Areas in the Seattle Area with Average Number of Hazards Identified per Structure

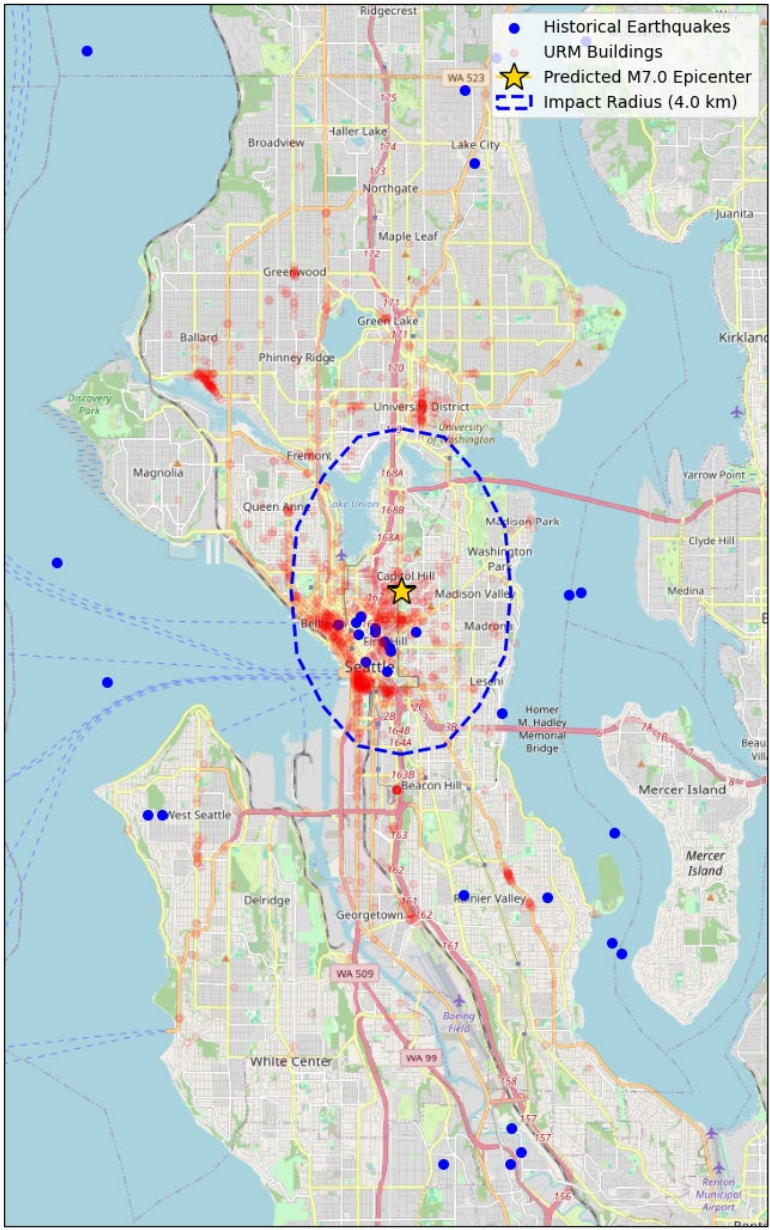


Analysis Maps #3 and #4:

Unreinforced Masonry Building Locations and Seattle Population Density Map

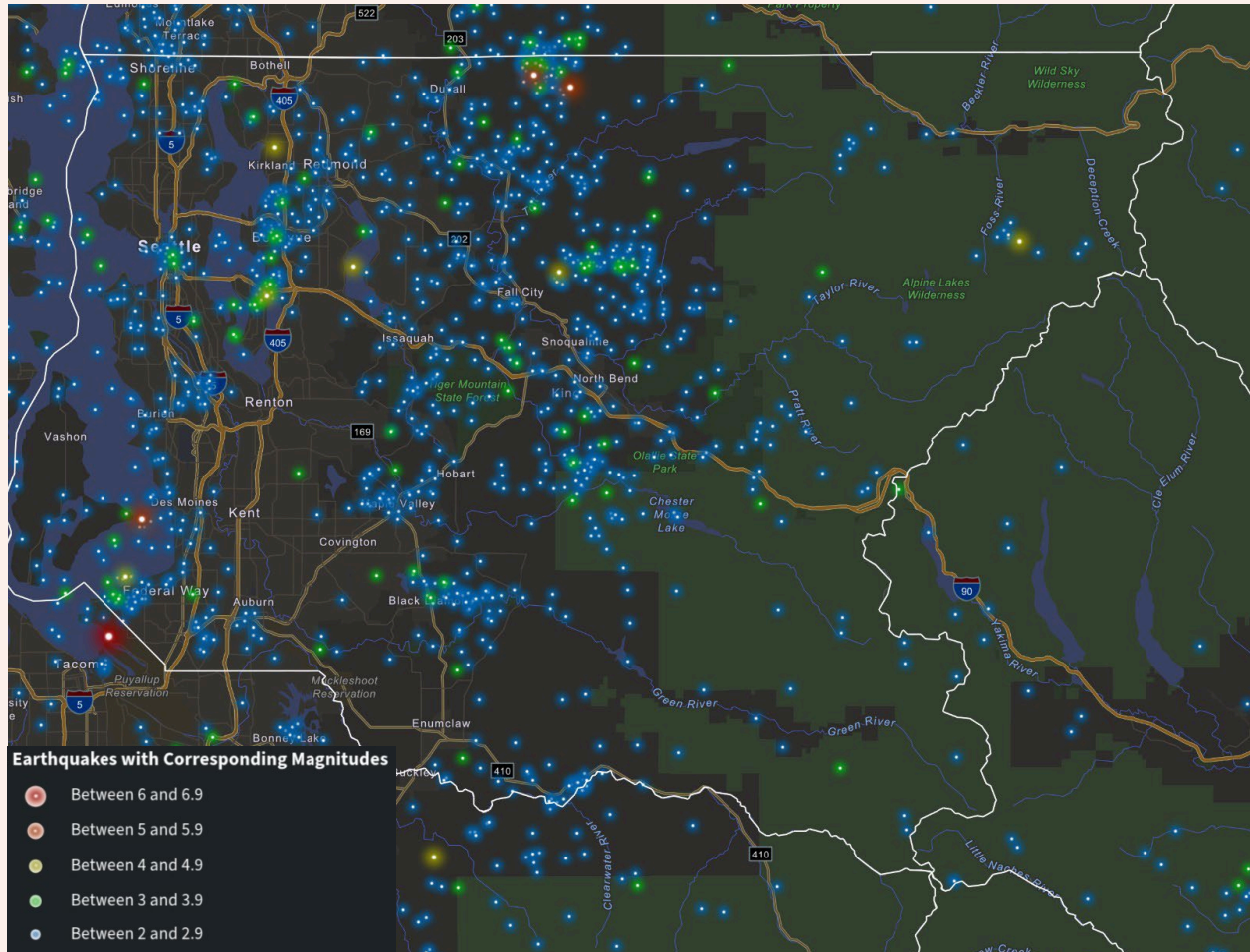


Predicted M7.0 Earthquake Impact in Seattle with URM Buildings

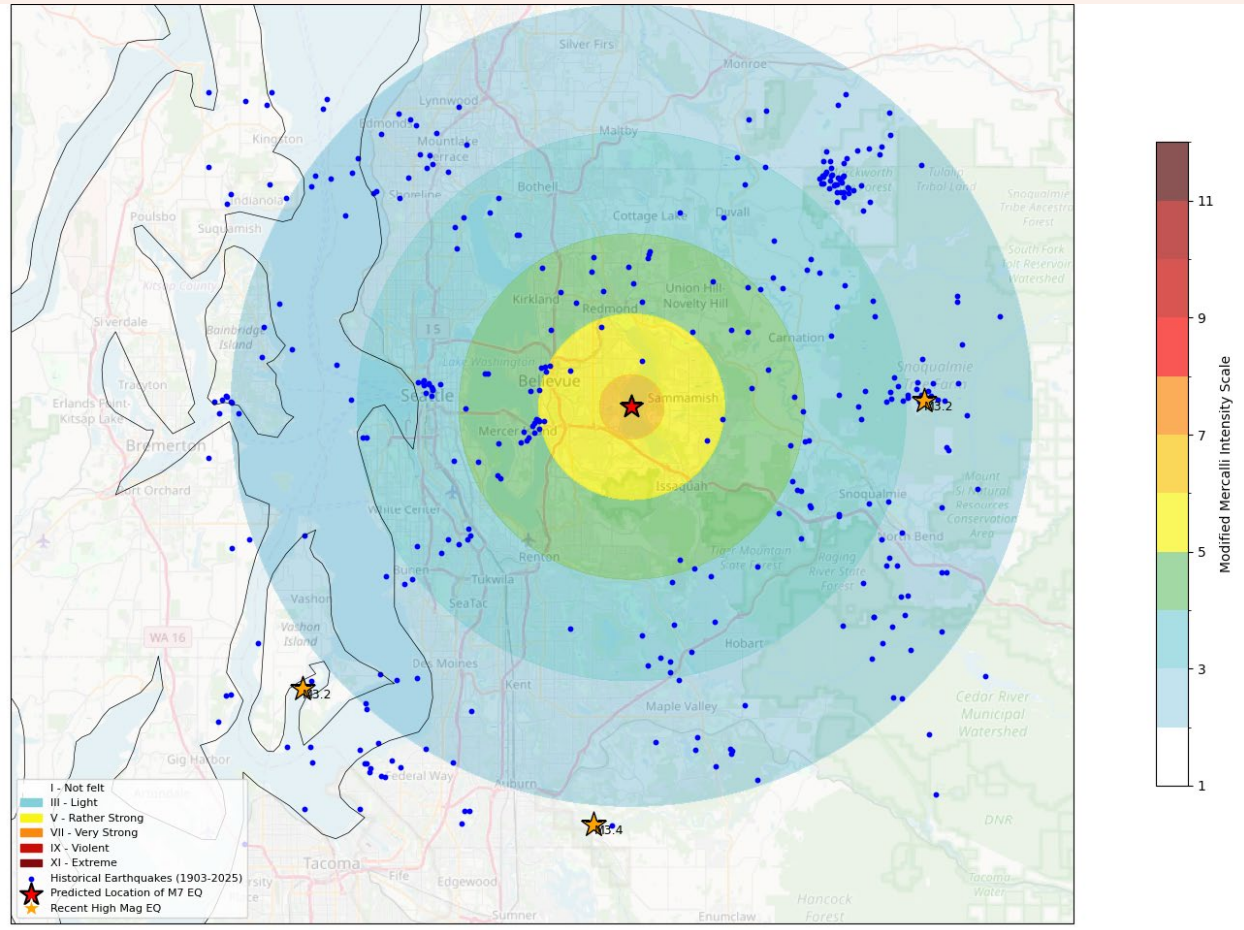


Analysis Map #5:

Compiled Earthquake Locations from 1900 to 2025 in King County

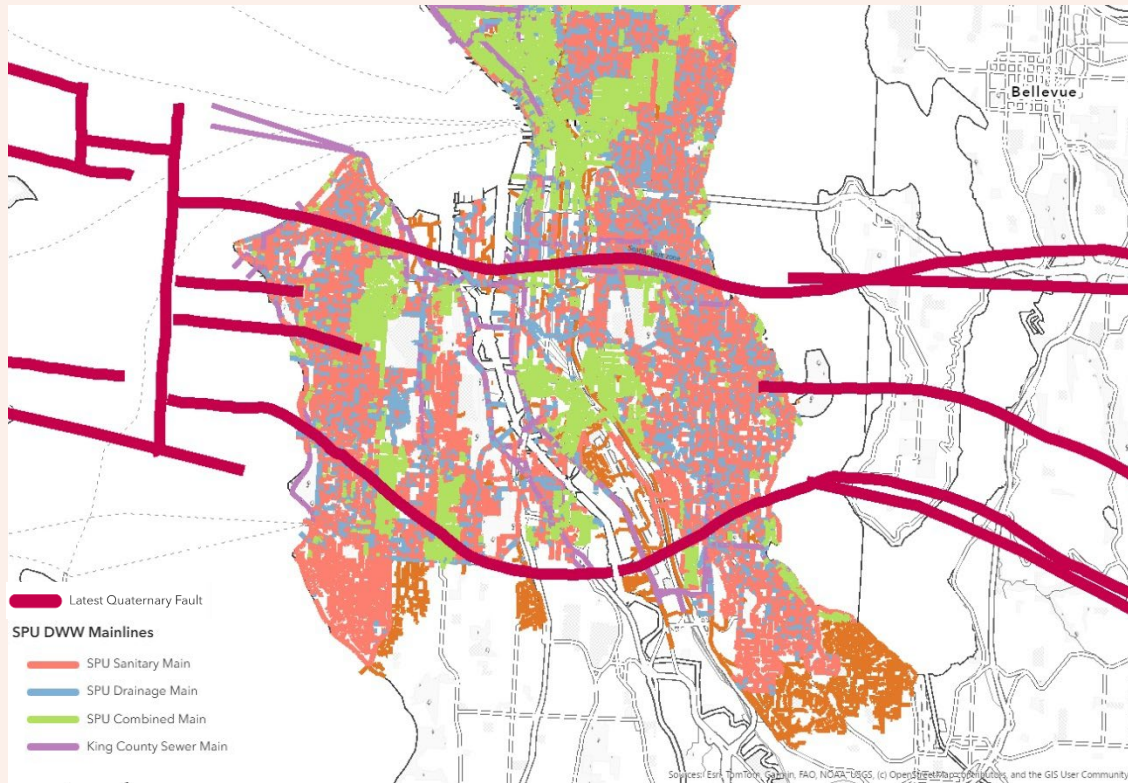


Analysis Map #6:
Modified Mercalli Intensity (MMI) Map
for Aftershocks of Simulated M7.0 Earthquake



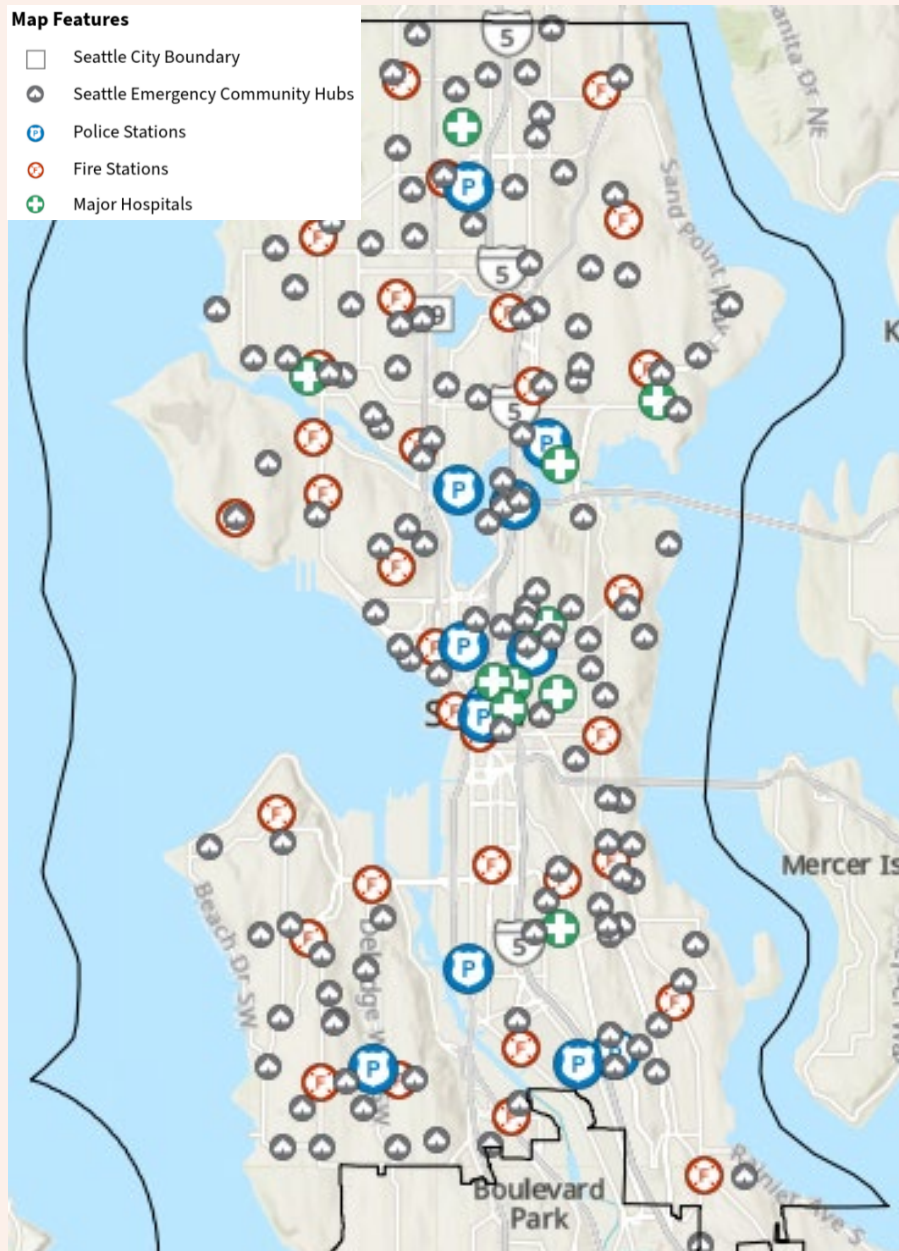
Analysis Map #7:

Overlap of Fault Lines Through Seattle and SPU DWW (Seattle Public Utilities Drainage and Wastewater) Main Lines



Analysis Map #8:

Emergency Services Locations and Community Emergency Hub Locations in Seattle



Written Analysis

This analysis examines how seismic activity in King County could impact bridges, transportation, flood-prone areas, and critical infrastructure. By mapping fault lines, liquefaction zones, and emergency services, it pinpoints key risks where there is a need for disaster planning.

The **first analysis map (Overlay of King County Seismic Bridge Risk Locations and Fault Lines in and Around Washington)** overlays the fault lines located in and around King County with the predicted levels of damage seismic activity would have on the bridges in the area. As we can see, several bridges that have a significant risk of damage from seismic activity are located on top of the fault lines, which have a moderate risk of causing an earthquake in the region. This map communicates hazard zones and evacuation routes by showing bridge damage risks (minor to severe in Seattle and Tacoma) and fault activity (historic and recent in Puget Sound), highlighting community impacts like disrupted I-5 and I-90 bridges during a seismic event.



This map also displays the many fault lines around the Seattle metropolitan area specifically. The Seattle Fault holds the greatest risk, running right through the land in the Puget Sound. A 7.5M earthquake would have many relations to this fault, likely causing MMI VII-VIII shaking in dense urban zones. Seattle's densest areas, with 10,000+ people, like downtown near the Seattle Fault Line, face heightened risk, complicating evacuation. Additionally, the secondary faults in and around this area intersect with much of King County's critical infrastructure, like I-5 and I-90, risking blockages during quakes. On-road vehicle evacuation would require using these very highways. The conditions of bridges like the I-90 over Mercer Slough will be called into question in the case of an earthquake. There will be some structural danger anywhere you try to go via car, making it difficult for people to evacuate from denser and more unstable liquefaction zones, especially in West and South Seattle. This is why an earthquake disaster is not one you evacuate from; instead, you stay put and take shelter in nearby areas. This data highlights exactly where these problems should be addressed, guiding retrofitting and shelter planning for the region's largest economic hub.

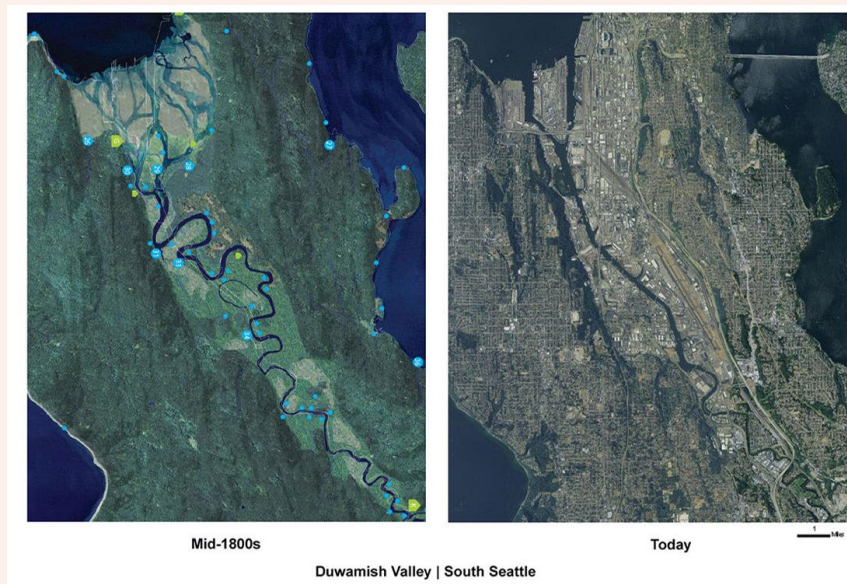
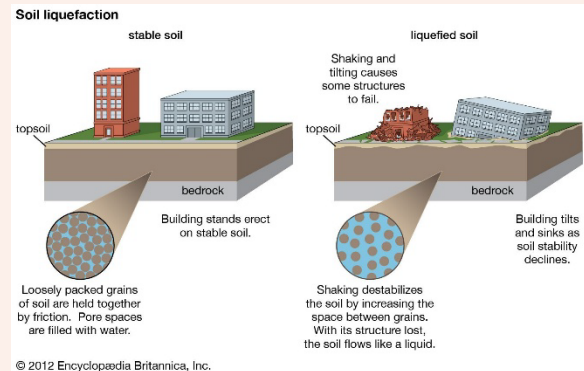


Figure 6 of the Collected Geospatial Data (Distribution of Land Use and Property Types in Liquefaction-Prone Areas of Seattle)

shows how Seattle utilizes its 531,738.37 acres. It also includes data on how much of its liquefaction-prone area is occupied by each specific land use category. Liquefaction areas can be very dangerous during an earthquake. This is what the percent of the zone represents. Covering about 9% of this land is commercial and mixed-use. Seattle is very

dense, and as we've also mentioned before, hospitals and critical infrastructure are vital during an earthquake. About 25% of Seattle's institutional land, including hospitals and police stations, is located inside the liquefaction-prone area. The right-of-way measures transportation and the road

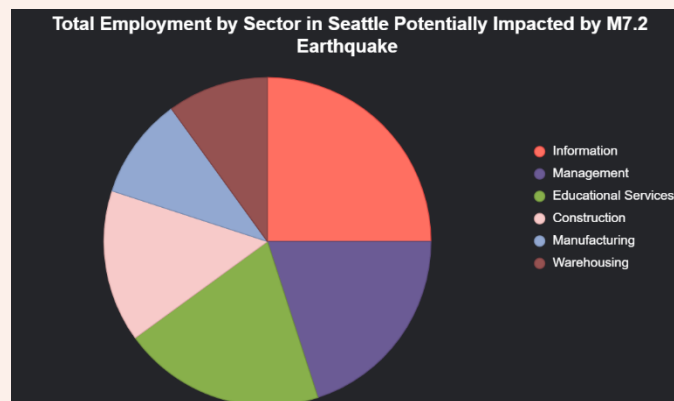
system in Seattle, which accounts for about 23% of the zone. This could block many inlets and exits out of Seattle, as we've mentioned in the past, since earthquakes can destroy transportation systems. Vehicle-based evacuations are difficult, and the dense commercial zones in downtown Seattle could also be a major threat. Seattle needs to retrofit its vulnerable areas with disaster preparations. Understanding the risk of land distribution is crucial before an earthquake.



A floodplain is a low-lying land mass that is adjacent to a river or stream. They are generally more prone to flooding during intense water flows. **The fourth figure of the Collected Geospatial Data (FEMA High-Risk Flood Area (100-year Floodplain) / FEMA floodplain 100-year area)** shows the regions of King County that are in a floodplain. Highlighting these areas is important because

earthquakes can make these areas hazardous due to soil liquefaction. Soil liquefaction is a term used to describe how wet, loose soil transforms into a liquefied state due to intense shaking during earthquakes. Because liquefaction can force water and sediment to the surface, it can potentially cause secondary events like flooding and landslides to occur. As a result, low-lying areas with a higher risk of flooding are more likely to be impacted by seismic activity. **Analysis map #2: Liquefaction & Flood Prone Areas in the Seattle Area with Average Number of Hazards Identified per Structure** shows specific areas in Seattle with their corresponding number of estimated hazard risks due to events like erosion, earthquakes, liquefaction, and flooding. The number of hazard risks per structure correlates with the high susceptibility liquefaction areas and flood prone areas of Seattle, as shown by the overlapping of white, pink, and blue regions. This map is helpful to understand the high-risk flood and liquefaction areas that people should potentially evacuate from in the event of a major earthquake.

Figure 13 of the collected geospatial data on the city of Seattle shows the projected employee impact by sector in a supposed M7.2 earthquake, which measures the intensity of an earthquake based on its observed effects on people, buildings, and the Earth's surface. In Seattle, due to the high concentration of big tech HQs here, the information, management, and educational services sectors have the highest employee exposure (over 200,000). Higher MMI levels (VII-VIII) show increased employee impact. In the construction, manufacturing, and warehousing sectors, there are also many strong employee impacts. This could be because of the large physical demand



of the space required to run these industries, and in an earthquake, many of these operations will not occur as the shaking amplifies risk. There's a definitive correlation between our supposed magnitude 7.2 earthquake and a potential magnitude 7.5 quake that occurs every 5,000 years in the Seattle area. This graph also shows the economic impact as an aftermath of the earthquake that may or may not happen. Seattle's major industries

suffer a lot of loss, for example, its main manufacturer, Boeing, which could experience destructive damage after a significant 7.2 earthquake. This graph shows which workers are most impacted and which industries will suffer a lot of damage. It could guide resource allocations before a massive earthquake that would result in less losses.

The Earthquake Magnitudes from 1900-2025 in King County map (Analysis Map #5) displays the intensity of the earthquakes based on the Richter scale in the map. These earthquakes have occurred between the 1900s and now. Most of the earthquakes in King County are between 2 and 3.9, suggesting that small earthquakes are very frequent. Even though these events do not cause significant damage, they emphasize that the area has underlying seismic activity. Earthquakes with magnitudes between 4 and 4.9 are scattered in clusters near fault lines. The

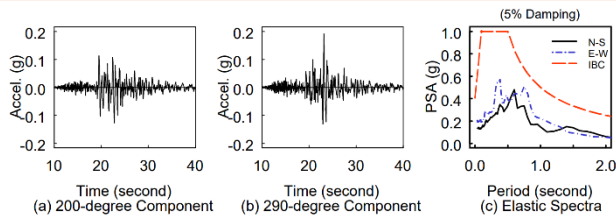


Figure 2.5 Seattle-Tacoma International Airport.

worst earthquakes that have magnitudes above 5 are rare on this map but highlight potential areas of stronger earthquakes, particularly in zones where soft soil or basin-edge effects increase ground motion. Past lethal earthquakes such as the 2001 Nisqually earthquake, caused lots of disruption that was widespread even though it was a deep-focus

earthquake. It damaged the traffic control tower at Seattle-Tacoma International Airport and affected many buildings, especially older structures. Older buildings were not built to the current seismic standards, increasing their vulnerability to earthquakes. The Nisqually earthquake alarmed emergency response agencies, causing many policy changes including enhanced building codes, targeting unreinforced buildings, and the utilization of advanced early warning systems like ShakeAlert. Data from magnitude graphs and historical events reinforce the necessity for regular maintenance measures like bracing shelves, water heaters, and major structural retrofits. Ultimately, geospatial analyses guide and project strategic investments in buildings and planning for emergencies, ensuring that communities remain safe, no matter if there is a small tremor or a significant earthquake.

After looking at past earthquakes like the one in Nisqually, we felt a need to predict the location of future earthquakes in the King County Area to better prepare for potential seismic events. Because earthquakes can happen anywhere and are very hard to predict, the exact location is impossible to determine. We took the most straightforward approach by averaging the epicenter coordinates of **all past 2.5M+ earthquakes in the area from 1900-2025 (data points from Figure 1 of the collected geospatial data)** to locate a possible spot.

```
# Calculate mean earthquake location (probable future earthquake spot)
mean_eq_loc = np.mean(eq_locs, axis=0)
print(f"Probable earthquake location: {mean_eq_loc[0]:.4f}°W, {mean_eq_loc[1]:.4f}°N")
```

While this was not for locating purposes, **Analysis Map #6** correctly shows the epicenter far away from major Pacific Northwest population centers based in Seattle. It uses an ETAS model to evaluate the spatial and temporal distribution of aftershocks given a simulated M7.0 earthquake. Aftershocks are smaller earthquakes that occur in the same general area during the days to years following a larger event. Many aftershocks are large enough to stop rescue services and destabilize buildings even more. The ETAS model marks those zones, which are of higher probability for aftershocks, so they can be given priority for retrofitting of critical utilities and buildings in the wider Seattle area. We use this piece of code to calculate the aftershocks' spatial impact.

```
def compute_etas_intensity(eq_times, eq_locs, eq_mags, current_time):
    lon_range = np.linspace(MAP_EXTENT[0], MAP_EXTENT[1], 200)
    lat_range = np.linspace(MAP_EXTENT[2], MAP_EXTENT[3], 200)
    lon_grid, lat_grid = np.meshgrid(lon_range, lat_range)
    mu, k, alpha, c, p, d, q, M0 = 0.01, 0.1, 1.0, 0.01, 1.0, 0.1, 2.0, 4.0
    _, _ = eq_probability = predict_earthquake_locations(eq_locs)
    intensity = np.full_like(lon_grid, mu) * eq_probability
    grid_points = np.stack([lon_grid.ravel(), lat_grid.ravel()], axis=1)
    for t_i, loc_i, M_i in zip(eq_times, eq_locs, eq_mags):
        if t_i < current_time:
            dt = current_time - t_i + c
```

```

r = np.sqrt(((grid_points[:, 0] - loc_i[0]) * 111)**2 + ((grid_points[:, 1] - loc_i[1]) * 111 *
np.cos(np.radians(loc_i[1]))**2) + d
productivity = k * np.exp(alpha * (M_i - M0))
temporal_term = productivity / (dt ** p)
spatial_term = 1 / (r ** q)
intensity += (temporal_term * spatial_term).reshape(lon_grid.shape)
return lon_grid, lat_grid, intensity

```

However, earthquake aftershocks are mostly measured in MMI as seen before when explaining employee impact. The intensity scale must be converted from the ETAS $\log_{10}$ form to MMI for understandability among professionals.

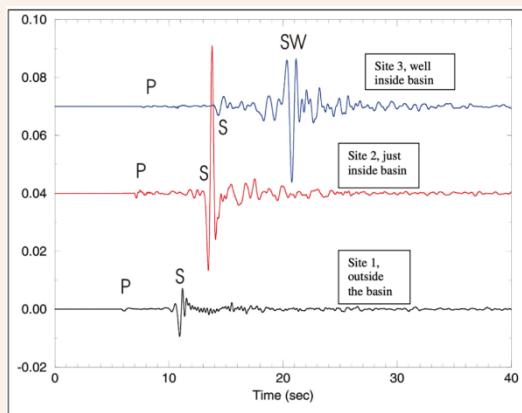
```

aftershock_mag = intensity
distances = np.sqrt(((lon_grid - mean_eq_loc[0]) * 111 * np.cos(np.radians(mean_eq_loc[1]))**2 + ((lat_grid -
mean_eq_loc[1]) * 111)**2)
log_pga = 0.3 + 0.59 * (aftershock_mag - 6) - 0.0075 * distances - 1.6 * np.log10(distances + 10)
pga = 10 ** log_pga
mmi = np.where(pga < 0.0017, 1.0, 3.66 * np.log10(pga * 980) - 1.66)
mmi = np.clip(mmi, 1, 12)
weighted_mmi = mmi * intensity / np.max(intensity) # Scale MMI by aftershock probability

```

After looking at the results, we can see that the intensity in Seattle is fairly moderate. This means the shaking, based on the weighted MMI, stays manageable even in crowded areas. However, aftershocks can happen at any time, and since an earthquake with the same magnitude as the 2001 Nisqually earthquake occurs about every 30 to 50 years, another quake or its aftershocks could happen even today.

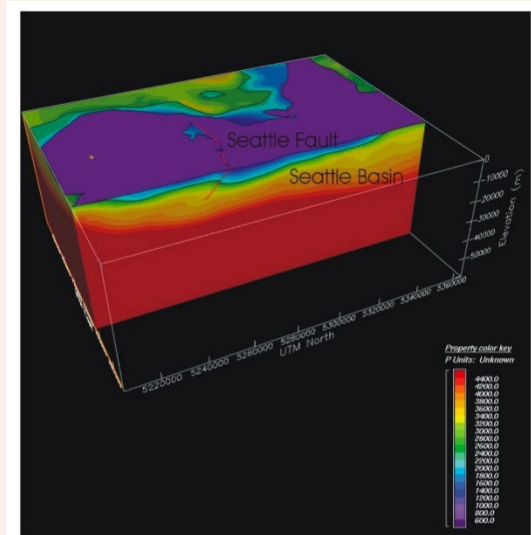
Another aspect that is quite important to the spatial analysis of earthquakes is the integration of three-dimensional (3D) sedimentary basin effects which help in modeling the phenomenon of earthquakes shaking more violently and for longer in places with soft water-filled sediments, like estuaries. The Seattle seismic array stations are on soft foundations; thus, they detect slower waves since the foundation is not hard enough. In contrast, Pacific Northwest Seismic Array stations are on solid bedrock; thus, they detect considerably faster waves. More surface materials, including loose surface materials such as alluvial deposits or glacial till, can dampen the waves and result in more intense shaking where there is a high concentration of soft sediments. The physical interactions of seismic waves that mainly consist of compressional waves and shear waves have geological structures that help us better understand the local earthquake impacts, especially in more vulnerable environments such as estuaries and coastal environments. Estuaries are often a part of larger sedimentary basins that have a thick, soft sediments that are water saturated. These materials have lower velocities compared to the bedrock that is

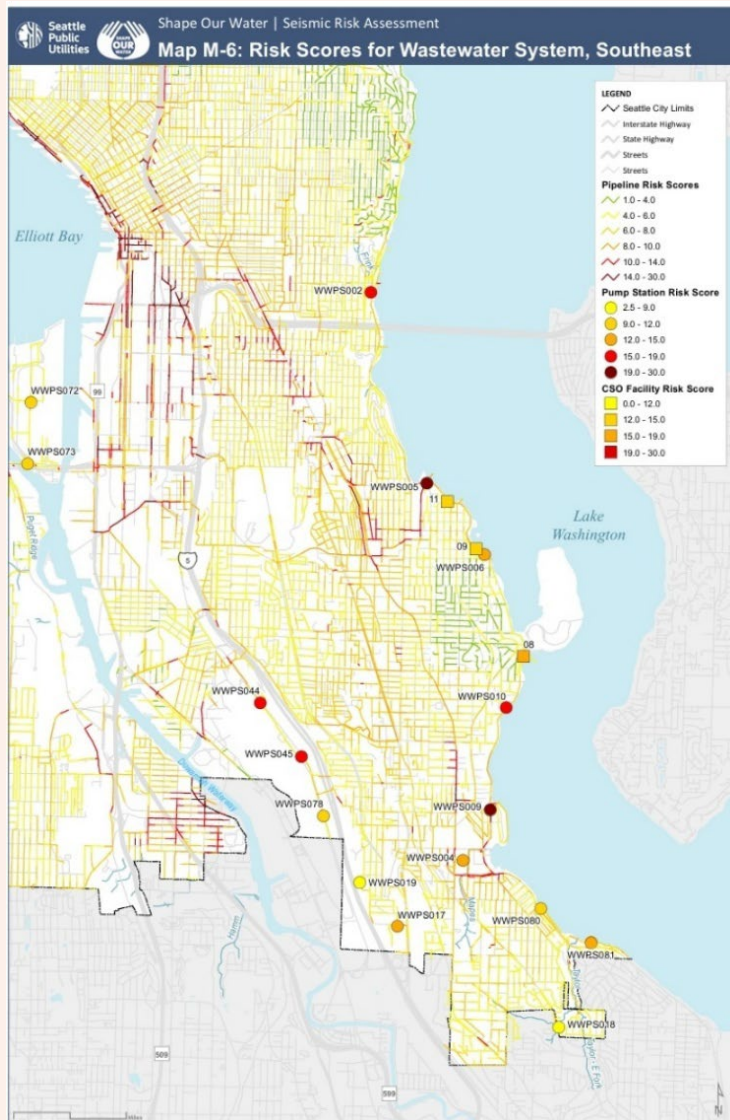


underneath. Compression waves travel faster and move through both solids and fluids by compressing the material, face the softer sediments, resulting in a decrease in the velocity. Meanwhile, shear waves are more damaging because they can displace material that is perpendicular to their direction of travel and they only move through solids, significantly decreasing the velocity. The decrease in wave speed for both types of waves and the decrease in wavelength leads to a significant increase in the amplitude. This amplification arises from the conservation of energy and the sharp

impedance contrasts at the boundaries between bedrock and the softer basin fill, a primary physical reason for the heightened shaking intensity observed in such areas. The main reason for the increased shaking intensity in such areas is the amplification in the increase of the amplitude of the waves due to the decrease in wavelength in order to conserve energy. The basin structure in estuarine settings traps both compressional and shear wave energy. This provokes wave reflections and refractions, increasing the duration of the shaking. The conversion of waves such as shear waves into slower moving and large amplitude waves move within the sediment layers, taking more control of the earthquake ground motion and causing a significant amount of structural damage. With Puget Sound being a major estuary, it is important to take this analysis into consideration.

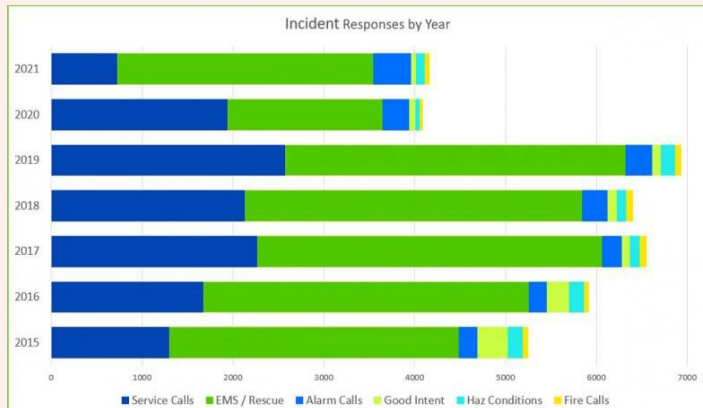
3D simulations model how earthquake shaking is affected by complex geological structures such as sedimentary basins. They begin by creating detailed geological 3D models of the subsurface. Two of its parameters or factors include the velocities of compressional wave velocity and shear wave velocity. These simulations showcase the movement of seismic energy during various earthquakes. This reveals the shape of the basin and explains how the sediments trap, focus, and convert seismic waves, leading to amplified and increased ground shaking in specific areas. This is how synthetic seismograms and amplification maps are created. Then they are compared to the recordings of actual earthquakes in order to refine these models.





Related to what goes on beneath the surface and what we see with our eyes during an earthquake are the problems that arise when you factor in almost 2000 miles worth of underground sewer and drainage pipelines running under the city. Wastewater systems are quite vulnerable during earthquakes because so much of it is underground, like mainlines, pump stations, storage tanks, and sewer systems that can all get damaged when the ground starts shaking. **In analysis map #7** we see exactly how faults surrounding the Puget Sound and King County areas crisscross through so many pipelines managed by Seattle Public Utilities. If the pipes break, shift, or the pump stations get damaged, it could cause sewage to back up into people's homes. In some cases, it could even leak untreated sewage into the environment. Due to Seattle's normally rainy climate, the runoff could easily carry this unwanted, toxic, and bacteria-filled sewage all over the surface of the city. This situation could expose people to harmful germs, raise the risk of water contamination, and possibly cause serious health

problems. We reviewed a study by Seattle Public Utilities and Brown and Caldwell that assessed earthquake risks to Seattle's wastewater systems and ranked them to help prioritize repairs and planning. (*Seattle Public Utilities* 36). The map to the left shows Southeast Seattle, where there is far more risk involved in the wastewater system's seismic vulnerability. This seems to be especially true near stations like WWP5004, which means this area might need some urgent upgrades or fixes to prevent issues during an earthquake. The conclusions drawn by the research study use many useful data points, from ground acceleration to the physical attributes of the pipelines, which are summed up to create the risk scores. This data almost entirely lines up with where the faults illustrated in map #7 run through. Where there is a fault, there is a likelihood of higher risk of damaged pipelines.

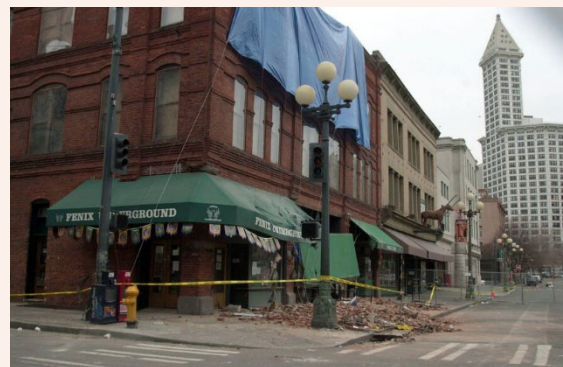


The Emergency Services Locations and Community Emergency Hub Locations in Seattle Map (Analysis Map # 8)

displays hospitals, police stations, fire stations, and community gathering locations in Seattle. Emergency managers can better prepare for earthquake scenarios by identifying nearby public safety facilities when a seismic event arrives. Such maps conveniently ensure that first responders

and other decision-makers can project the focus toward areas with a higher need of assistance. Additionally, residents can gather in community hubs to help each other, share information, and access resources before outside help arrives. This above graph also highlights potential service gaps. The graph delineates incident responses by year across King County, showcasing the distribution of service calls, EMS/rescue, alarm calls, good intent, hazardous conditions, and fire calls to aid emergency planning and resource allocation. Officials can target areas with fewer hospitals or shelters for additional resources. Additionally, tracking older buildings prone to collapse or areas with unstable soil alongside these important sites can make planners prioritize structural upgrades. The Federal Emergency Management Agency emphasizes the importance of geospatial data in creating robust plans to prepare for earthquakes. Communities significantly benefit from comprehensive hazard maps when funding for protective measures. Similarly, the Washington Geological Survey emphasizes the importance of identifying local faults and weak soils to protect hospitals, utility lanes, etc. Along with guidelines from the US Geological Survey, maps of facilities are very powerful planning tools. Visually combining resource locations with seismic risk data allows communities to better prepare themselves for future earthquakes, protect lives, and lower the number of disruptions to everyday life.

URM stands for Unreinforced Masonry buildings, which are old brick buildings that were typically built prior to 1945. There is a reported 1164 URMs throughout the city of Seattle. And according to seattle.gov, “URMs are vulnerable to damage and collapse during earthquakes because the parapets and walls are not secured to the roofs and floors. Parapets can break away, falling into the street and putting pedestrians in danger. Walls can break away and lead to full or partial collapse.” Knowing the locations of these buildings is important to understand the areas in Seattle where people should take extra precautions in case of seismic activity. Taking account of the **seventh map of our Collected Geospatial Data (1160 Unreinforced Masonry Buildings in Seattle | City of Seattle)**, the analysis map created (**3: Unreinforced Building Risk X Seattle Population**) shows the



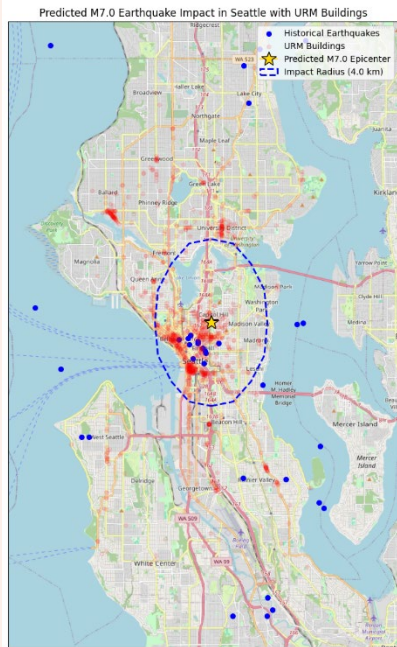
locations of these URMs on top of a population density map of Seattle. Areas with a higher population density as well as URM nearby should be mapped out for emergency services, because in the event of a major earthquake there would likely severely damage the URMs and create more casualties in these locations.

Analysis Map #4 (Predicted M7.0 Earthquake Impact in Seattle with URM buildings) was created using a Python program to average past earthquake locations and simulate a new M7.0 earthquake epicenter, including an impact radius calculated from the area in km^2 using the formula $\text{Area (Km}^2\text{)} = 10^{0.5M-1.8}$, and overlays this data with unreinforced masonry buildings to identify high-risk zones in Seattle.

```
#part of the code
avg_lat = seattle_eq['latitude'].mean()
avg_lon = seattle_eq['longitude'].mean()
pred_mag = 7.0
pred_epicenter = (avg_lon, avg_lat)
impact_area = 10 ** (0.5 * pred_mag - 1.8)
impact_radius = np.sqrt(impact_area / np.pi)
print(f"Predicted epicenter: lat={avg_lat:.4f}, lon={avg_lon:.4f}")
print(f"Impact area: {impact_area:.2f} km^2, Impact radius: {impact_radius:.2f} km")
```

This script builds collected geospatial and the provided map image, using libraries like pandas, numpy, matplotlib, and cartopy for geospatial plotting.

Team Prediction



According to our analysis of the maps and the map above, **we predict that the most dangerous place in Seattle will be around URM (Unreinforced Masonry Buildings), as they have the highest risk of collapsing.** Citizens should be ready to evacuate immediately from these buildings after experiencing an earthquake and emergency responders should be prepared to head to these areas first. **We also predict that since several overpasses (on major roads) and bridges lie on top of several secondary faults, they will become hazardous and unsafe to travel on during and after seismic activity.** Because of this, we believe that the evacuation routes should follow inner roads to avoid these structures. Additionally, **liquefaction prone areas in the urban, suburban, and metropolitan areas can become susceptible to flooding (especially areas that overlap with low-lying areas or floodplains),** so local response agencies such as fire departments and police stations should head to these areas to potentially aid in evacuations.

In contrast to the Nisqually Earthquake of 2001, **we predict that a major earthquake happening in Seattle would cause a much greater economic impact due to Seattle's high population density.** The epicenter of the Nisqually Earthquake occurred near Olympia which had a population of around 60,000 people and caused around \$4 billion in property damage. However, Seattle has a population of around 755,000! Due to our findings that Seattle contains high concentrations of big tech HQs, and with the information, management, and educational services sectors having the highest employee exposure (over 200,000), **we predict that an earthquake with a magnitude greater than 6M along the Seattle Fault will cause at least \$50 billion dollars in damage if not more.**

Based on our analysis of map 7, **we also conclude that where there is a fault, there is a likelihood of higher risk of damaged pipelines.** Sewage managements teams should stand ready in case toxic untreated sewage gets leaked into the environment or backed up in residents' homes.

Earthquakes are unique natural disasters because scientists are still not able to accurately predict when they will happen. The exact timing, location, or magnitude of an earthquake doesn't become known unless it has already occurred. With this and our geospatial analysis in mind, **we've come up with the conclusion that the best thing to do in an earthquake is to not evacuate from Seattle but stay put.** If someone is inside a structurally sound building in the event of a major earthquake, they should expect to not be the priority of emergency first responders, as locations with unreinforced masonry buildings are more likely to be damaged and sought after. Staying indoors would ensure that they stay safe if any aftershocks were to occur. If someone is outside,

they should stay away from buildings (specifically URMs that would be prone to collapsing), and should avoid bridges or overpasses that cross over dangerous faults. Finally, if someone is driving, they should go to the side of the road, stop, and wait until the shaking stops. However, if someone is driving on a bridge or an overpass, our data proves that it is much safer to continue driving than to stop on an elevated road.

There is huge potential for widespread damage and not much our community can do during this disaster. **Planning ahead of time by using key research and geospatial data to focus on hazard mitigation (reinforcing buildings and developing early warning systems) will be the best way to minimize the impact.** The question isn't if Seattle will face a major earthquake, but when. So, as we assess these risks, the real challenge awaits: Will we take action before it's too late?

This concludes our predictions on what would happen if an earthquake on the Seattle Fault were to occur, as well as how the entire community should respond to this type of disaster.

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Work Log

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time Involved	Team member responsible (student initials)	Comments
11/28/24	Figure out a disaster threat to do geospatial analysis on and assign roles.	2 hrs	CB,HA,AP	We first picked wildfires, only to realize it is not viable, hence we chose earthquakes. CB and HA collected geospatial data, AP worked on drafting written analysis from the collected data.
12/5/24	Find good databases where we can get GIS data from and start making maps.	1 hr	CB,HA	CB and HA found databases from local and national sources such as King County GIS, USGS, and FEMA.
12/15/24	CB and HA to finish 2 maps each.	2.5 hrs	CB,HA	CB completed FEMA High Risk Flood Area and 2 Puget Sound Faults. HA finished Flood Risk Map and Bridges Maintained by King County.
1/3/2025	CB to start the portfolio and to input everything the team has.	1 hr	CB	CB started the portfolio and set the theme. Additionally, CB added data, maps, and explanations from before.
1/5/2025	HA and AP to do the Challenge explanation and the Challenge interpretation respectively.	1 hr	AP,HA	HA and AP finished the challenge explanations, interpretations, and gave three rounds of refinements or adding more information.
1/11/25	CB and HA to finish 2 maps each and to do more research.	2.5 hrs	CB,HA	CB and HA finished Analysis Map #1: Overlay of Fault Lines and Bridges

1/18/25	Team meeting everyone start checking off everything for Jan deadline.	45 min	CB,HA,AP	The materials have been revised, maps were finished, and analysis was completed. They need to be placed in the correct order, planned to be finished in 1/22/2025.
1/22/25	Team meeting to order all Geospatial Data and Analysis Maps in allocated spaces in portfolio and finish it.	2 hrs	CB,HA,AP	Geospatial data and analysis maps have been put in the correct locations. They are submitted and there is a plan to add more analysis maps and analysis paragraphs.
2/19/25	CB finishes 4 paragraphs of the analysis. HA finish 2. AP finish 2. CB to edit and refine.	3-4 hrs	CB,AP	CB, HA, and AP have finished paragraphs of analysis, including paragraphs upon map analysis. CB made refinements.
2/22/25	AP to finish the Prediction working with team during meeting.	3 hrs	CB,HA,AP	AP first wrote a draft of prediction. Then CB and HA provided refinements.
3/9/25	Finalize portfolio.	3 hrs	CB,HA,AP	The portfolio had formatting issues fixed by CB. HA and AP finalized and completed on paragraphs.
3/10/25	1 st day for Infographic.	3 hrs	CB,AP	CB and AP added the maps and text into the infographic. AP created a template/structure which will define the design. We also defined the color schemes.
3/13/25	Finish Infographic and submit portfolio.	2.5 hrs	CB,AP,HA	We all had one last meeting to add finishing touches to portfolio and AP got the draft of the infographic done.

5/14/25	Python Program and touching up analysis maps 2 and adding legends	3 hrs	CB	CB works on Python scripts that visualize earthquakes while AP created better legends for all the analysis maps as well as redid analysis map #2
5/17/25	CB and HA meet to find 6 more maps that could be added to collected geospatial data	3 hrs	CB, HA	CB and HA worked on adding maps and geospatial analysis
5/19/25	CB's Python Scripts	3 hrs	CB	CB finishes Python scripts that visualize earthquakes
5/19/25	HA starts explanation of physics and wave functions	1 hr	HA	HA worked on explaining the 3D models related to physic and seismic wave functions.
5/20/25	HA finishes explanation of physics	2 hr	HA	Added graphs spatial data related to ground acceleration and wave functions.
5/20/25	CB writes explanation of new geospatial data. AP writes final prediction. AP also adds captions for the collected geospatial data.	3 hr	AP, CB	CB added data points related to explanation. AP organizes the doc making sure that everything has a source and is properly defined. AP reviews the entire doc and bolds all references to maps as well as the predictions.
5/21/25 - Nationals	AP will create Nationals Infographic		AP	AP will update and add refinements to the State Infographic to prepare it for the TSA National Conference

Forms

STUDENT COPYRIGHT CHECKLIST (for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO

If NO, go to question 2.

If YES, is the music and/or sound copyrighted? ☐ YES ☒ NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.

1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, [Signature] (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.

2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, [Signature] (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, [Signature] (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Name: Cleofus Balaranjith

Chapter Advisor Signature: [Signature]